

YEAR 12

- 1) A (-1; 6), B, C (3; 6) and D are the vertices of a square, A B C D. B and D both lie on the line $x = 1$. Determine the coordinates of B and D.
- 2) Which of the following describes the movement of a figure that is translated according to the rule below?

$$(x, y) \rightarrow (x - 7, y + 1)$$

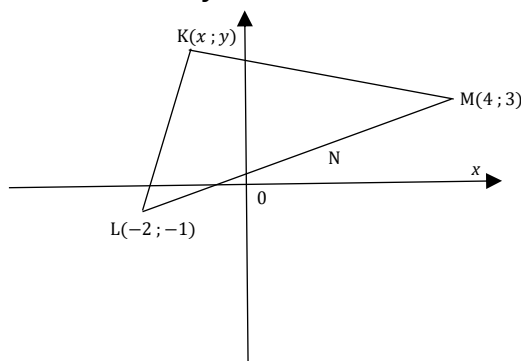
- F. down 7 units and right 1 unit.
- G. left 7 units and up 1 unit.
- H. right 7 units and down 1 unit.
- J. up 7 units and left 1 unit.

A B C D is a parallelogram, where points A (2; 5), B (5; 3) and C (0; r) are given. Point D lies in the second quadrant. (Question 3-8)

- 3) If the gradient of BC is $\frac{4}{0}$, calculate the value of r
- 4) Calculate the coordinate of D
- 5) Calculate the size of angle B $\hat{C}$ D rounded off to 2 decimal places.
- 6) Determine the length of BC
- 7) Determine the equation of the line perpendicular to BC, passing through point A.
- 8) Determine the area of parallelogram A B C D.
- 9) Prove that the straight line $2y - 3x = -25$ is a tangent to the circle $x^2 + y^2 - 4x + 6y = 0$ at B (5 ; - 5)

In the accompanying figure, K (x ; y), L (-2; -1) and M (4; 3) are the vertices of triangle K L M. the equation of the side KL is $y - 5x - 9 = 0$ and that of KM is $5y + x - 19 = 0$

N is the midpoint of LM
(Question 10-15)



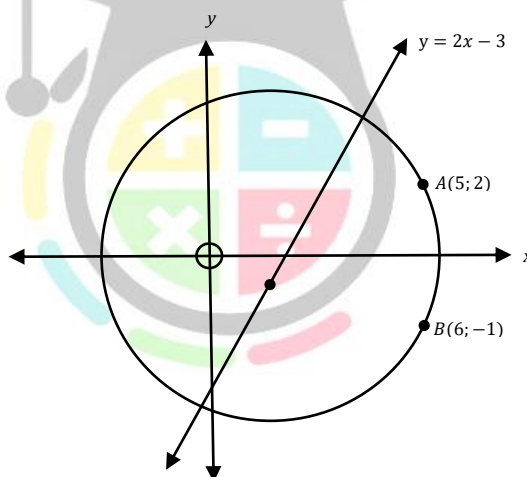
- 10) Calculate the coordinates of N
- 11) Show that the coordinates of K are (-1; 4)
- 12) Determine the equation of the line through K and N in the form $y = mx + c$

- 13) Determine the gradient of the line LM, hence, prove that KN is the perpendicular bisector of LM
- 14) If L, M and the point J (7; a) are collinear, calculate the value of a
- 15) Determine the size of the angle of inclination between KL and the positive x-axis

Quadrilateral ABCD is made up by the points A (8; a), B (6; 5), C (3; 6) and D (2; 3) (Question 16-18)

- 16) If $AB = AD$, calculate the value of a
- 17) Prove: $AC \perp BD$
- 18) Determine whether quadrilateral ABCD is a kite. Provide reasons for your answer.

The points A (5; 2) and B (6; -1) lie on a circle where the Centre of the circle lies on the line $y = 2x - 3$



- 19) Find the coordinates of the Centre of the circle.
- 20) Find the equation of the tangent line passing through the point A.